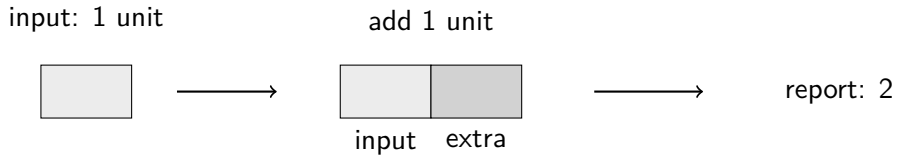
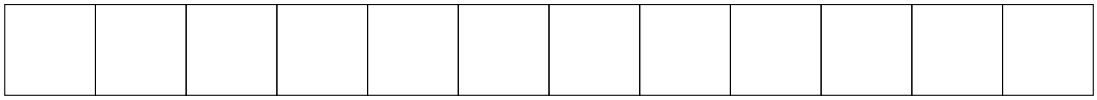
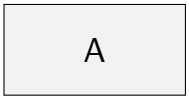
A machine reports a strip's length plus a secret extra number: 1, 2, 3, 4, 5, or 6. The extra number stays the same for the whole round. All lengths are whole units. Each square in the paper row is one unit.



Problem 1: Play the machine. Secretly choose an extra number. Your partner gives you a strip; tell them its report. They may ask for more strips until they can name your extra number. Reveal it and switch jobs.

Round	Strips requested	Reports
1		
2		
3		
4		











Problem 2: Two hidden strips give these reports. Find every possible pair of lengths.

P reports 7	Q reports 10

Length of P	Length of Q	Extra number

Problem 3: Could P be longer than Q? What can you know about the two lengths without learning the extra number?

Problem 4: Each row uses a different machine. A known strip gives report 8. Two hidden strips give reports 9 and 12. Build the hidden strips for each row.

Known strip	Length that reports 9	Length that reports 12
4 units $\longrightarrow$ 8		
6 units $\longrightarrow$ 8		
2 units $\longrightarrow$ 8		

Problem 5: Your partner knows only that two hidden strips report 7 and 10. Secretly choose a machine and write both hidden lengths under a cover. Let your partner request the report for one known strip, then build each hidden strip. Reveal the lengths and switch jobs.